Ranked Biomedical Future-Direction Recommendation

Tensor Rank Fusion of Knowledge-Graph, Entity & LLM Signals, with Per-Direction GNFP Analysis — Paper Plan & To-Do

Draft for Nourah & Naz — 2026-07-09

Paper Plan & To-Do — Ranked Biomedical Future-Direction Recommendation

Working title: *From List to Ranked List: Tensor Rank Fusion of Knowledge-Graph, Entity, and LLM Signals for Biomedical Future-Direction Recommendation, with Per-Direction Gap–Novelty–Feasibility–Publishability Analysis.*

One-sentence contribution. Given candidate future research directions extracted from a PubMed article, we (1) **rank** them by fusing Semantic Scholar knowledge-graph signals, biomedical entity relevance, and multiple LLM-agent judgments through a **context-conditioned low-rank tensor**, calibrated to *realized* future directions (forward citations); and (2) attach to each direction a **Gap / Novelty / Feasibility / Publishability (GNFP)** score *validated against later research outcomes*, turning a flat list into a decision-support ranking.

Positioning (defines the paper's identity)

- **Nourah's system (upstream, separate paper):** an agentic AI pipeline that *generates* a list of candidate future directions from PubMed articles — the **candidate generator**.
- **This paper (downstream):** takes that list (as one source among several) and contributes **ranking + fusion + per-direction GNFP analysis**. The interface is the candidate pool $\mathcal{C}$; the two papers must stay cleanly separable (no salami-slicing).

Status: tensor method designed and validated on synthetic + one real ranker (general/ML domain). This plan **ports it to biomedicine/PubMed**, adds the KG/entity/LLM sources, and adds the GNFP contribution.

> **Honest framing (put this in the paper).** No single primitive is new — CP/Tucker, CMTF, factorization machines, LLM-ranker ensembling, scientometric novelty scores, MRR selection all exist. The originality is the **intersection**: (a) *ranking* biomedical future directions against a **temporally-grounded realized-direction ground truth**; (b) a **tensor fusion of KG + entity + LLM signals** calibrated to that metric; (c) an **outcome-validated GNFP layer**. The ranking contribution stands only if the tensor beats weighted RRF **and** a learned meta-ranker with identical features (gate C2); the GNFP contribution stands only if its scores predict realized outcomes better than baselines (gate C5).

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0. Task, data, and ground truth (biomedical)

Task. Input = a PubMed article (body / abstract / conclusions, with explicit future-work sentences masked). Output = a **ranked** list of K candidate future-research directions, **each annotated with GNFP scores**. The system never sees the answer at inference.

Candidate pool $\mathcal{C}$ — Nourah's agentic generator is source (a): - (a) **Agentic LLM generation** (Nourah's system) — primary generative source. - (b) **Retrieval** — kNN over a *future-work / conclusions bank* built from training-era

PubMed articles (SPECTER2 / PubMedBERT / BioLinkBERT embeddings; optional BM25 hybrid). - (c) **Citation-grounded** — sentences from forward-citing papers that state realized next steps.

Ground truth (two label sources on a temporal split). 1. **Masked future-work statements** — atomic-split into directions $g_1, \ldots, g_m$. 2. **Forward-citation realized directions** — directions later papers *actually pursued*, detected via (entity-combination first-appearance or semantic match) in citing literature after time T. A time-grounded, outcome-based label the general-ML setting lacked.

Temporal split (no leakage). Train/dev on articles $\leq T_{\text{train}}$; test on the most recent. Banks, embeddings, KG snapshots, entity co-occurrence counts, and all tuned parameters use $\leq T$ data only. Select on dev, report once on test.

MRR match function on free text. match$(d_i, g_j) = 1$ if sim$(d_i, g_j) \geq \tau$ (cosine over SPECTER2 / PubMedBERT), **or** a biomedical entailment judge says "covers," **or** the key entity set of d_i is a superset of that of g_j. Report judge–human agreement (Cohen's κ); the judge only matches, never ranks. Sweep τ.

Data & tools. Corpus: **PubMed / MEDLINE** + **PubMed Central OA** full text. Graph: **Semantic Scholar Academic Graph (S2AG)** — citations, co-citation, bibliographic coupling, influential-citation counts, SPECTER2 embeddings. Entities/KG: **PubTator Central / MeSH / UMLS / scispaCy**; **SemMedDB / Hetionet** as the biomedical relation graph.

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1. Ranking signals = the tensor modes

Source m	Ranks candidates by	Backing
`llm_gpt`, `llm_claude`, `llm_biomed`	LLM-agent judgment of "is this the paper's next step"	GPT, Claude, a biomedical LLM (BioGPT / PubMedBERT reranker)
`s2_graph`	KG position: citation/co-citation/bib-coupling proximity + embedding similarity of the direction's topic to the paper's S2 neighborhood	S2AG
`entity_rel`	biomedical **entity relevance**: overlap / KG-proximity between the direction's entities and the paper's; centrality & emergence of those entities	PubTator + SemMedDB/Hetionet
`retrieval`	kNN distance to the future-work bank	SPECTER2 / PubMedBERT
`horizon`	recency / popularity / novelty priors from the citation graph	S2AG (§3.5)

Each source m produces a ranking $r_m(d \mid p)$ of the shared pool $\mathcal{C}$.

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2. The tensor rank-fusion model (complete formulation)

This is the originality anchor. It is written to be lifted directly into the paper's Method section.

2.0 Notation

Symbol	Meaning
$P,\ D,\ M$	no. of papers, candidate directions (per paper), sources
$F,\ q$	no. of paper-context features, candidate side-info columns
$R,\ R'$	CP / context-weight rank, and interaction (FM) rank
$r_m(d \mid p)$	rank of candidate d from source m for paper p (∞ if unranked)
k_m	per-source RRF smoothing constant
$\phi_m(d \mid p)$	reciprocal-rank signal $1/(k_m + r_m)$ (0 if unranked)
$\mathcal{X}$	order-3 fusion tensor, $\mathcal{X}[p, d, m] = \phi_m(d \mid p)$, shape $P \times D \times M$
$A,\ B,\ C$	CP factor matrices (paper / candidate / source): $P \times R,\ D \times R,\ M \times R$
$z(p)$	paper-context feature vector, $\in \mathbb{R}^F$
$U,\ w_m(p)$	context→source-weight map $M \times F$; per-paper weight of source m
$G_s,\ G_c$	low-rank factors of $U = G_s G_c^\top$: $M \times R,\ F \times R$
v_m	FM interaction embedding of source m, $\in \mathbb{R}^{R'}$
$H,\ W$	candidate side-info matrix $D \times q$ and its factor $q \times R$ ($H \approx BW^\top$)

2.1 The object

For paper p, candidate d, source m, the reciprocal-rank signal

$$\phi_m(d \mid p) = \frac{1}{k_m + r_m(d \mid p)} \qquad (= 0 \text{ if source } m \text{ did not rank } d \text{ for } p).$$

Stack into the **order-3 tensor** $\mathcal{X} \in \mathbb{R}^{P \times D \times M}$, $\mathcal{X}[p, d, m] = \phi_m(d \mid p)$. A **fiber** $\mathcal{X}[p, d, :]$ is a candidate's "source profile"; a **slice** $\mathcal{X}[p, :, :]$ is paper p's source-by-candidate matrix. **Rank fusion is an operation on the mode-M (source) direction of this tensor.**

2.2 Tensor-algebra preliminaries (needed for fitting)

- **Rank-1 tensor / outer product:** $a \circ b \circ c$ has entries $a_i b_j c_k$.
- **Mode-n unfolding (matricization):** $\mathbf{X}_{(n)}$ reshapes the tensor into a matrix whose columns are its mode-n fibers, with $\mathbf{X}_{(1)} \in \mathbb{R}^{P \times DM}$, $\mathbf{X}_{(2)} \in \mathbb{R}^{D \times PM}$, $\mathbf{X}_{(3)} \in \mathbb{R}^{M \times PD}$.
- **Mode-n product $\mathcal{X} \times_n V$:** contract mode n with matrix/vector V.
- **Khatri-Rao product** $\odot$ (column-wise Kronecker): the algebra that turns a CP model into a matrix equation:

$$\mathbf{X}_{(1)} = A\,(C \odot B)^\top, \qquad \mathbf{X}_{(2)} = B\,(C \odot A)^\top, \qquad \mathbf{X}_{(3)} = C\,(B \odot A)^\top.$$

  These three identities are exactly what the ALS updates in §2.8 solve.

2.3 Plain weighted RRF, in tensor language

Weighted RRF is a **mode-3 product** with a fixed source-weight vector $w \in \mathbb{R}^M$:

$$s(\cdot \mid p) = \mathbf{X}[p, :, :]\, w \qquad \Longleftrightarrow \qquad S = \mathcal{X} \times_3 w^\top.$$

The entire model family below removes one assumption at a time from this line: *stop collapsing mode M with a single, global, frozen vector.*

2.4 CP / PARAFAC: the consensus view

Canonical Polyadic decomposition approximates the tensor by R rank-1 terms:

$$\mathcal{X} \approx \sum_{r=1}^{R} \lambda_r \, (a_r \circ b_r \circ c_r), \qquad \hat{\mathcal{X}}[p, d, m] = \sum_r \lambda_r \, A[p, r] \, B[d, r] \, C[m, r].$$

- Each r is a latent **consensus axis** (e.g. "LLM-agreement," "KG+entity," "citation-momentum"). - $C[m, r]$ = **how much source** m **loads on axis** r — an interpretable source-trust readout. - The fused score is the low-rank reconstruction $s(d \mid p) = \sum_m \hat{\mathcal{X}}[p, d, m]$, which **denoises** idiosyncratic single-source errors and **rewards structure shared across sources**. - **Kruskal uniqueness:** under the mild condition $k_A + k_B + k_C \geq 2R + 2$ (k = Kruskal rank), the CP factors are unique up to scaling and permutation — so the consensus axes are *identifiable*, not arbitrary rotations (unlike PCA/matrix factorization). This is the theoretical basis of the interpretability story. - **Transductivity catch:** the paper factor $A[p, :]$ is indexed by p; it is fit per training paper and is undefined for a new test paper. Pure CP therefore cannot score an unseen paper. §2.5 fixes this by tying the paper mode to features.

2.5 The deployed model: context-conditioned low-rank fusion (inductive)

Replace the free per-paper factor with a linear function of paper-context features $z(p) \in \mathbb{R}^F$:

$$a_p := \Theta \, z(p) \qquad (\Theta \in \mathbb{R}^{R \times F}) \qquad \text{— the paper factor becomes a function, not a lookup.}$$

Folding Θ and the candidate mode into the source weights yields a **per-source weight that adapts to the paper**:

$$w_m(p) = \langle U_m, \, z(p) \rangle, \qquad U \in \mathbb{R}^{M \times F}, \quad U = G_s G_c^{\top}, \;\; \text{rank}(U) = R,$$

and the **first-order (bilinear) score**

$$s_1(d \mid p) = \sum_m w_m(p) \, \phi_m(d \mid p) = \sum_m \langle U_m, \, z(p) \rangle \, \phi_m(d \mid p).$$

This is a low-rank tensor coupling (source $\times$ context $\times$ rank-signal). Read the limiting cases:

- U **constant across** z (only a bias column active) $\Rightarrow w_m(p) = w_m \Rightarrow$ **exactly weighted RRF**. RRF is a strict special case.
- U **full rank** ($R = \min(M, F)$) $\Rightarrow$ unconstrained query-dependent weighting; overfits when papers are few.
- **small** R **(1–3)** $\Rightarrow$ shares an R-dim latent structure across sources *and* contexts; the regime we deploy.

Full deployed scorer = first-order + interaction + optional linear bias:

$$\begin{aligned} s(d \mid p) = & \sum_m \langle U_m, \, z(p) \rangle \, \phi_m(d \mid p) && \text{(A) context-adaptive per-source trust} \\ & + \sum_{m < m'} \langle v_m, \, v_{m'} \rangle \, \phi_m(d \mid p) \, \phi_{m'}(d \mid p) && \text{(B) low-rank interaction (FM)} \\ & + \; b_0 + \langle \beta, \, x(d, p) \rangle && \text{(C) bias / candidate features} \end{aligned}$$

- **(B)** makes "GPT and Claude *agreeing*" or "KG *confirms* the entity signal" a **learned multiplicative** term (a symmetric rank-R' tensor, i.e. a Factorization Machine), rather than a hand-crafted `gpt_claude_rank_corr` feature.
- **(C)** couples the candidate mode to side information — see §2.6.

2.6 Coupling to side information (CMTF) — the distinctive piece

The horizon priors + entity/KG features form a candidate side-info matrix $H \in \mathbb{R}^{D \times q}$. Two ways to inject it, both sharing the **candidate factor** B so KG/entity signals live in the *same latent space* as the ranking consensus:

Route 1 — factorization-first (unsupervised, then read out). Jointly factorize the tensor and the matrix (classic CMTF, Acar et al.):

$$\min_{A,B,C,W} \left\|\mathcal{X} - [\![A,B,C]\!]\right\|_F^2 \;+\; \mu\left\|H - BW^\top\right\|_F^2 \;+\; \Omega(A,B,C,W).$$

Then fuse by low-rank reconstruction and regress A on $z(p)$ for induction.

Route 2 — end-to-end discriminative (deployed). Train the scorer $s(d \mid p)$ of §2.5 directly by a ranking loss (§2.7), and add the coupling as a **representation regularizer** that ties the candidate embeddings B (which generate the β / interaction terms for d) to H:

$$\mathcal{L}_{\text{couple}} = \mu\left\|H - BW^\top\right\|_F^2.$$

Route 2 is what we deploy (it optimizes the metric-aligned loss directly); Route 1 is the baseline/ablation and a clean initialization for B. **State both; report the ablation with vs without $\mathcal{L}_{\text{couple}}$ (gate C3).**

2.7 Training objectives (MRR is non-differentiable, use surrogates)

Labels: within paper p, candidate d is **positive** if it matches a realized direction (§0). Train the *scores* $s(d \mid p)$ with an LTR surrogate, **select everything by true dev MRR.**

Pairwise logistic (RankNet). Over positive–negative pairs (d^+, d^-) within a paper:

$$\mathcal{L}_{\text{pair}} = \sum_{p} \sum_{(d^+,d^-)} \log\Big(1 + \exp(-(s(d^+ \mid p) - s(d^- \mid p)))\Big).$$

Listwise (ListNet / softmax cross-entropy). With a top-1 target distribution from the labels:

$$\mathcal{L}_{\text{list}} = -\sum_{p}\sum_{d} y(d \mid p)\, \log \text{softmax}_d(s(d \mid p)), \qquad y = \text{normalized relevance}.$$

Full regularized objective (deployed, Route 2):

$$\min_{U,v,\beta,b_0,W} \;\mathcal{L}_{\text{rank}}(s) \;+\; \mu\left\|H - BW^\top\right\|_F^2 \;+\; \alpha\Big(\|G_s\|^2 + \|G_c\|^2 + \textstyle\sum_m \|v_m\|^2 + \|\beta\|^2\Big).$$

Here $\mathcal{L}_{\text{rank}} \in \{\mathcal{L}_{\text{pair}}, \mathcal{L}_{\text{list}}\}$. Hyperparameters $\{R, R', \mu, \alpha, k_m\}$ and the loss choice are chosen by dev MRR — the loss is a differentiable proxy; **the metric decides.**

2.8 Optimization

ALS for CP / CMTF (Route 1). Cyclically fix two factors and solve the third in closed form from the matricized equations of §2.2 ($*$ = Hadamard product, $^+$ = pseudo-inverse):

$$A \leftarrow \mathbf{X}_{(1)}(C \odot B)\big((C^\top C) * (B^\top B)\big)^+$$
$$B \leftarrow \big[\,\mathbf{X}_{(2)}(C \odot A)\;;\; \sqrt{\mu}\,HW\,\big] \text{ solved jointly (candidate factor sees both } \mathcal{X} \text{ and } H)$$
$$C \leftarrow \mathbf{X}_{(3)}(B \odot A)\big((B^\top B) * (A^\top A)\big)^+$$
$$W \leftarrow (B^\top B)^+ B^\top H$$

Each update is a small $R \times R$ linear solve. Alternatively **CMTF-OPT** (Acar): all-at-once gradient on (A,B,C,W) — more stable under degeneracy.

Adam for the discriminative scorer (Route 2). Back-propagate $\mathcal{L}_{\text{rank}} + \mathcal{L}_{\text{couple}} + \text{reg}$ through $s(d \mid p)$. Gradients are cheap and sparse (only observed $\phi_m \neq 0$ contribute); e.g.

$$\frac{\partial s(d \mid p)}{\partial U_m} = \phi_m(d \mid p)\, z(p)^\top, \qquad \frac{\partial s(d \mid p)}{\partial v_m} = \phi_m(d \mid p) \sum_{m' \neq m} \langle v_{m'}, \cdot \rangle\, \phi_{m'}(d \mid p) \text{ (standard FM gradient).}$$

Init & scaling. HOSVD or random init for CP; initialize B from Route-1 CMTF; absorb column norms into λ_r; optional non-negativity ($A, B, C \geq 0$) for cleaner, parts-based consensus axes.

2.9 Rank & hyperparameter selection

Grid or Bayesian (Optuna/TPE) over $\{R, R', \mu, \alpha, k_m\}$; early-stop on dev MRR (not on the surrogate loss). Report the MRR-vs-R curve — the paper's evidence that low rank is doing the regularizing.

2.10 Missing sources & cold start (why the tensor is well-posed here)

- **Missing entries.** $\phi_m(d \mid p) = 0$ encodes "source m did not rank d." Fit with a **masked loss over observed (ranked) entries** so zeros are not treated as strong negatives; the low-rank structure *imputes* the missing source's opinion — a built-in advantage over RRF, which simply drops the term.
- **New paper at inference (inductive).** Score with $z(p)$ via §2.5 — no a_p lookup needed. This is exactly why we abandoned pure CP.
- **New candidate.** Represented by its source profile $\mathcal{X}[p, d, :]$ plus its side-info row $H[d, :]$; no per-candidate parameter is required.

2.11 Complexity

- **ALS / Adam epoch:** $O(\text{nnz}(\mathcal{X}) \cdot R)$ where $\text{nnz}(\mathcal{X}) \leq P \cdot D \cdot M$ (typically 5–20% dense). Milliseconds at hackathon scale.
- **Inference per paper:** $O(D{\cdot}M{\cdot}R + D{\cdot}R'^2)$ — a few matrix-vector products. **Cheaper than the LightGBM meta-ranker**, and far cheaper than any additional LLM call.

2.12 Identifiability, degeneracy, and when NOT to use it

- **Identifiability:** Kruskal's condition (§2.4) makes the axes meaningful — cite it as the reason the interpretability claims are valid.
- **Degeneracy:** CP can diverge (two factors growing to cancel). Mitigate with ridge α, non-negativity, or orthogonality on C; prefer CMTF-OPT over vanilla ALS if it appears.
- **Honest failure mode:** with M small (~5–7) and *no* real cross-source or context structure, the fitted U collapses to constant and the model *is* RRF. If dev MRR of the tensor $\approx$ RRF, **say so and keep RRF** — the tensor only earns the paper when structure exists (gate C2). CP/Tucker at higher order is the scale-up path only when M grows (many LLMs $\times$ prompts $\times$ embeddings).

2.13 Worked micro-example (intuition)

Two papers, three sources, one candidate d. Suppose on **method-heavy** papers `entity_rel` is trustworthy and `llm_*` noisy, and on **clinical** papers it is the reverse. Global RRF must pick one w and is wrong half the time. The context model learns $w_{\text{entity}}(p) = \langle U_{\text{entity}}, z(p) \rangle$ that is high when $z(p)$ says "method-heavy" and low when "clinical" — a single rank-1 factor ($R = 1$) already captures this flip that *no* global weight vector can. That is the whole pitch in one example.

2.14 Parameter-count / capability comparison

$P = 300,\ D = 100,\ M = 7,\ F = 8,\ R = 2,\ R' = 2,\ q = 6$:

Model	Params	Generalizes to new paper?	Interpretable source-trust?
Weighted RRF	7 (+7 k)	yes	global only
Query-dependent weights (full U)	56	yes	yes, but overfits
Context tensor (A), $R = 2$	30	yes	**yes, per-paper**
+ interaction (B), $R' = 2$	+14	yes	yes
+ CMTF coupling (C)	$+2q = +12$	yes	yes + KG/horizon coupled
LightGBM meta-ranker	100s–1000s (trees)	yes	via SHAP, post-hoc

Pitch: **fewer parameters than the GBDT, adapts per paper, weights directly readable** — but earns the paper only if it lifts MRR over both baselines (gate C2).

2.15 Algorithm (deployed, Route 2)

```
Input: rankings {r_m(d|p)}, context z(p), side-info H, labels y; ranks R,R'; mu, alpha
1.  Build X[p,d,m] = 1/(k_m + r_m(d|p)); mark observed mask Omega
2.  Init B via Route-1 CMTF (few ALS sweeps on X, H); init G_s,G_c,v,beta ~ small random
3.  repeat until dev-MRR stalls:
4.      for each minibatch of papers:
5.          s(d|p) = sum_m <U_m,z(p)> phi_m + sum_{m<m'} <v_m,v_m'> phi_m phi_m' + b0 + <beta,x>
6.          L = L_rank(s, y over mask Omega) + mu*||H - B W^T||_F^2 + alpha*reg
7.          Adam step on (G_s, G_c, v, beta, b0, W)
8.      evaluate dev MRR; keep best checkpoint
9.  return best (by dev MRR), plus loadings C / w_m(p) for explanations
```

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3. Per-direction GNFP analysis (second contribution)

Each surfaced direction ships four scores, **each computed only from information available at time T and validated against realized outcomes at $T + \Delta$** (temporal split → a genuine *prediction* task, not a subjective rubric).

3.1 The four scores (computed at time T)

- **Gap(d)** — under-exploration of d's core entity combination:

$$\text{Gap}(d) = \text{adjacency}(d, p) \cdot (-\log(1 + \text{cooccur}_{\leq T}(\text{entities}(d)))),$$

  adjacency-gated by $\cos(e_d, e_{\text{paper}})$ so off-topic $\neq$ "gap."
- **Novelty(d)** — embedding distance of d from the paper's prior-art neighborhood at T + first-co-mention indicator of its entity combination.
- **Feasibility(d)** — *softest axis, be explicit.* Proxies at T: existence of datasets/assays/methods linking d's entities (method/data edges in the KG), resource intensity, entity "tractability." An LLM-judge **calibrated against a small expert-labeled set** + KG proxies; report agreement.
- **Publishability(d)** — a scientometric-momentum model predicting whether pursuing d yields a published, cited result, trained on historical (direction → outcome) pairs.

3.2 Outcome labels (observable at $T + \Delta$, held out)

Pursued? (a later paper pursued d) · **Time-to-pursuit** (months) · **Impact-if-pursued** (citations of the pursuing papers).

3.3 Validation (what makes GNFP publishable)

On the temporal test set, report **AUC / correlation of each GNFP score against its realized outcome** — does high Feasibility predict shorter time-to-pursuit? does high Gap $\times$ Novelty predict higher impact-if-pursued? Beat baselines (raw citation count, LLM-only score, random). If GNFP does not beat baselines, **downgrade to a qualitative analysis and say so** (gate C5).

3.4 Aggregation to a priority score

Weighted **geometric** mean (conjunctive: a fatal-low axis tanks the score), weights **fit to maximize correlation with realized impact**:

$$\text{Priority}(d) = \exp\Big(\sum_i w_i \ \ln \text{GNFP}_i(d)\Big).$$

Optional EV/utility form $U = P(\text{pursue}) \cdot \text{Impact}/\text{Effort}$ (reuses the project's Compute-Risk Governor). Report **rank stability** via Monte-Carlo over score uncertainty.

3.5 How GNFP and the tensor connect

- GNFP axes become **context/side-info features** feeding the tensor ($z(p)$ and H).
- The tensor's **consensus-axis loadings** feed the *explanation* of each GNFP score.
- Compare two ranked lists: **MRR-optimal** (predict what the field *did*) vs **GNFP-priority** (what a researcher *should* pursue); their divergence is itself a finding.
- *Restraint:* the load-bearing tensor is the **rank fusion**; GNFP is a calibrated, outcome-validated scoring layer, not a second forced tensor.

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4. Claims to prove (empirical spine)

- **C1 (ranking):** tensor fusion (A) $>$ weighted RRF on test MRR, significant margin.
- **C2 (decisive):** tensor (A / A+B / A+B+C) $\geq$ LightGBM/logistic meta-ranker **with identical features**. *If C2 fails, re-headline to the biomedical benchmark + GNFP.*
- **C3 (side-info):** CMTF coupling (C) adds marginal MRR over the plain tensor.
- **C4 (entity signal):** ablating `entity_rel` / `s2_graph` measurably drops MRR (justifies the biomedical KG framing).
- **C5 (GNFP validity):** each GNFP score predicts its realized outcome better than baselines.
- **C6 (interpretability):** consensus-axis loadings + per-paper source weights are faithful (match leave-one-signal-out attribution) and readable.
- **C7 (robustness):** results hold across MeSH areas / length / no. of GT directions and across the τ sweep.

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5. Evaluation design (summary)

- **Primary:** MRR (+ MRR@5/10, Recall@K, nDCG@K, MAP) vs realized directions, temporal test.
- **Candidate-recall ceiling table first** (rankers can't fix a missing candidate).
- **GNFP validation table** (§3.3): per-axis AUC/correlation vs outcomes.
- **Ablations** (decisive rows): single-source $\rightarrow$ RRF $\rightarrow$ meta-ranker $\rightarrow$ tensor A $\rightarrow$ +B $\rightarrow$ +C; plus source-dropout (entity, KG, each LLM).
- **Significance:** bootstrap CIs, paired tests. **Leakage audit:** all banks/counts/embeddings from $\leq T$; out-of-fold feature generation.

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6. To-Do roadmap to publication

Phase 0 — De-risk & align · *~2–3 days*

- ☐ **Coordinate with Nourah**: fix the candidate-pool interface (schema, per-article $\mathcal{C}$); agree the two papers' boundaries in writing.
- ☐ Synthetic sanity check: does tensor (A) beat RRF + logistic meta-ranker when the generator injects *context-dependent, entity-correlated* source reliability (the §2.13 flip)? Fix the model if not.
- ☐ Freeze the MRR match function (τ, entailment judge, entity-overlap rule) and the temporal split.

Phase 1 — Biomedical data (the real bottleneck) · *~2 weeks*

- ☐ Build the **temporal-split PubMed/PMC corpus** (articles with explicit future-work/conclusions).
- ☐ Entity-link with PubTator/MeSH/UMLS; snapshot the KG (S2AG + SemMedDB/Hetionet) at $\leq T$.
- ☐ Generate per-article sources: Nourah's directions, retrieval bank, citation candidates, `s2_graph`, `entity_rel`, `horizon`, the 3 LLM rankers.
- ☐ Derive **realized-direction labels** from forward citations. Leakage audit; candidate-recall ceiling table.

Phase 2 — Ranking method & ablations · *~1 week*

- ☐ Implement `tensor_fusion.py` (variants A / A+B / A+B+C; Route-1 CMTF init + Route-2 Adam) extending `rank_fusion/`.
- ☐ Baselines: single-source, tuned weighted RRF, logistic + LightGBM meta-ranker.
- ☐ Fit (pairwise-logistic / ListNet); select $\{R, R', \mu, \alpha, k_m\}$ by dev MRR; MRR-vs-R curve.
- ☐ Full ablation table incl. the decisive tensor-vs-meta-ranker rows and source-dropout.

Phase 3 — GNFP layer · *~1–1.5 weeks*

- ☐ Implement Gap / Novelty / Feasibility / Publishability scorers (KG proxies + calibrated LLM-judge for Feasibility).
- ☐ Build outcome labels at $T + \Delta$; validation table (per-axis AUC/correlation vs baselines).
- ☐ Fit priority-score weights on dev outcomes; Monte-Carlo rank stability.

Phase 4 — Interpretability & stats · *~3–4 days*

- ☐ Consensus-axis naming from $C[m, r]$; per-paper $w_m(p)$ case studies; faithfulness rate.
- ☐ Calibration (ECE), abstention; bootstrap CIs, paired tests; τ sensitivity; MeSH-area slices.

Phase 5 — Writing · *~1 week*

- ☐ Related work vs: CP/Tucker (Kolda–Bader), CMTF/CGTF (Acar; arXiv:1809.08353), FM (Rendle), LLM-Blender (2306.02561), RRF (Cormack 2009), RankGPT/PRP, FutureGen (2503.16561), SPECTER2/PubMedBERT/BioLinkBERT, PubTator/SemMedDB/Hetionet.
- ☐ Method (§2), GNFP (§3), Experiments (§4–5), honest Limitations (small-N, feasibility-axis softness, primitives-not-new, KG-snapshot noise).
- ☐ Reproducibility appendix: prompts, splits, KG snapshot date, hyperparameter grids, seeds.

Phase 6 — Venue & submission · *ongoing*

- ☐ Target: **BioNLP (ACL)**, ***Findings***, **JBI / JAMIA / Bioinformatics** (applied), or **SDP @ ACL**. Elevate the methods claim only if C2 holds.
- ☐ Anonymization, artifact/code-release policy, page limits; optional arXiv preprint.

Cross-cutting risk gates

- ☐ **C2 gate:** tensor must beat the GBDT with same features, else re-headline.

- ☐ **C5 gate:** GNFP must predict outcomes, else demote to qualitative.
- ☐ **Feasibility-axis gate:** if the softest axis has no signal, drop it and present GNP.
- ☐ **Data-starvation gate:** keep R small (1–3) at biomedical scale; don't claim full CP/Tucker.
- ☐ **Overlap gate with Nourah:** if the boundary blurs, merge or re-split — do not publish two slices of one system.
- ☐ **Competition scan** before submission.

References

1. Kolda & Bader, *Tensor Decompositions and Applications*, SIAM Review, 2009.
2. Acar, Kolda & Dunlavy, *All-at-once Optimization for Coupled Matrix and Tensor Factorizations* (CMTF-OPT); *Coupled Graphs and Tensor Factorization for Recommender Systems*, arXiv:1809.08353.
3. Rendle, *Factorization Machines*, ICDM 2010.
4. Cormack, Clarke & Büttcher, *Reciprocal Rank Fusion*, SIGIR 2009.
5. Jiang et al., *LLM-Blender*, ACL 2023 (arXiv:2306.02561).
6. Sun et al., *RankGPT*; Qin et al., *Pairwise Ranking Prompting (PRP)*, 2023.
7. FutureGen, *LLM-RAG Approach to Generate the Future Work of a Scientific Article*, 2025 (arXiv:2503.16561).
8. Cohan et al., *SPECTER* / SPECTER2; Gu et al., *PubMedBERT*; *BioLinkBERT*.
9. Wei et al., *PubTator Central*; Kilicoglu et al., *SemMedDB*; Himmelstein et al., *Hetionet*.
10. Cao et al., *ListNet* 2007; Burges, *RankNet* / *LambdaRank*.